Grile arbori binari și generalizați

Fie X un nod intr-un arbore binar. X are 2 copii. Fie Y succesorul lui X in parcurgerea in inordine. Care din afirmatiile urmatoare este adevarata?

Select one:

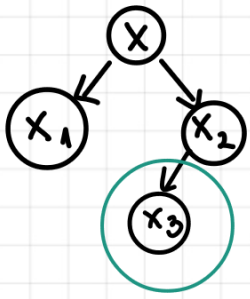
a. Y nu are copil in stanga

b. Y nu are copil in dreapta

c. Nici un raspuns nu este corect

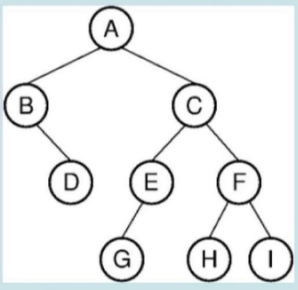
d. Y are ambii copii

parcurgere în inordine : stânga , rădăcină, dreapta



dacă ar exista asta, Y nu ar mai fi succesor

Sa se afiseze urmatorul arbore prin parcurgerea in preordine:



Se vor afisa literele ca majuscule separate prin virgula fara spatii:

Exemplu:

A,B,C

Answer: A,B,D,C,E,G,F,H,I

preordine - rădăcină, preordină (R Sb)

A, B, b, C, E, G, F, H, i

Sa se scrie configuratia pe niveluri a arborelui binar ordonat rezultat dupa insertia intr-un arbore vid a cheilor urmatoare in ordinea data mai jos:

10,7,6,8,9

Pentru fiecare nivel (de sus in jos) se vor scrie cheile (de la stanga la dreapta) separate prin virgula (,), iar nivelurile vor fi separate prin bara verticala (|) fara spatii.

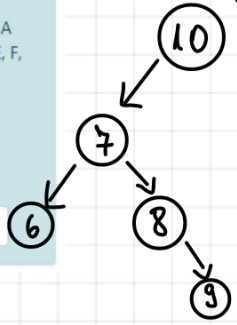
Exemplu A|B,C|D,E,F,G - reprezinta un arbore cu nodul A pe primul nivel, nodurile B si C pe al doilea nivel si D, E, F, G pe al treilea nivel.

Answer: 10|7|6,8|9

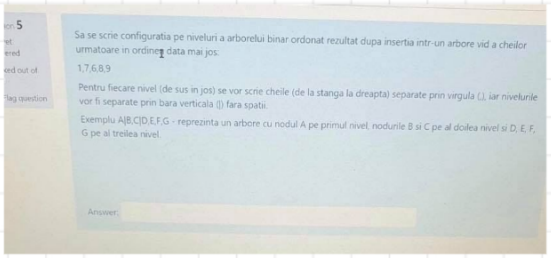
→ la inordine, va ieși crescător

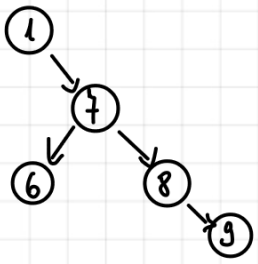
→ stânga (cheie mai mică)

→ dreapta (cheie mai mare)

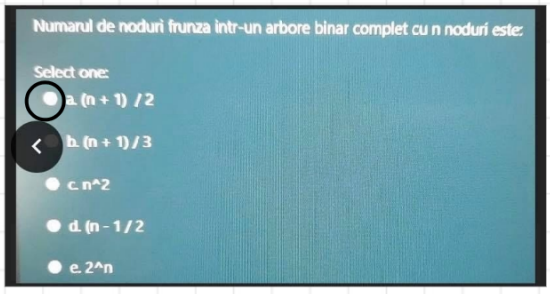


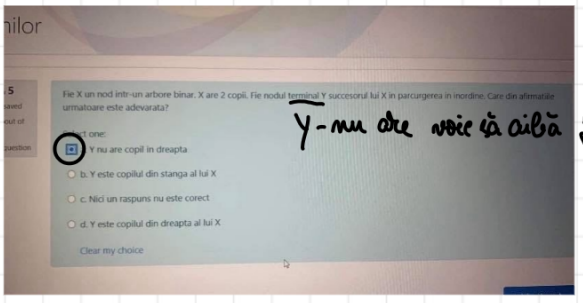
10|7|6,8|9



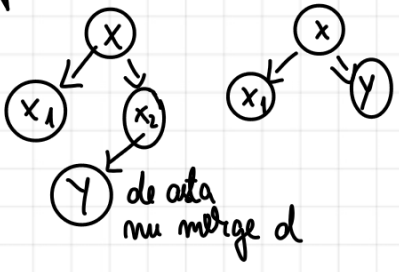


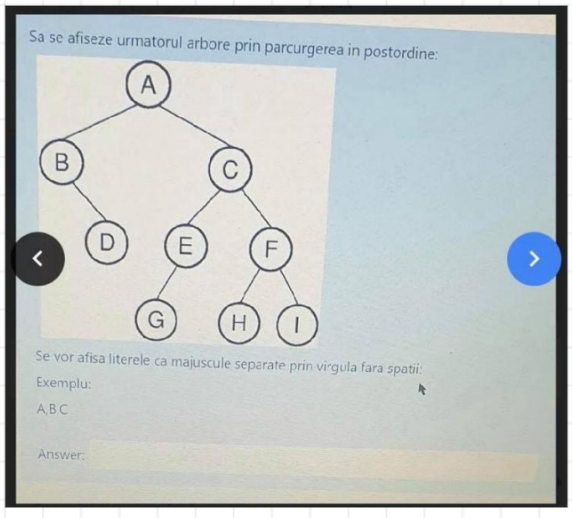
1|7|6,8|9





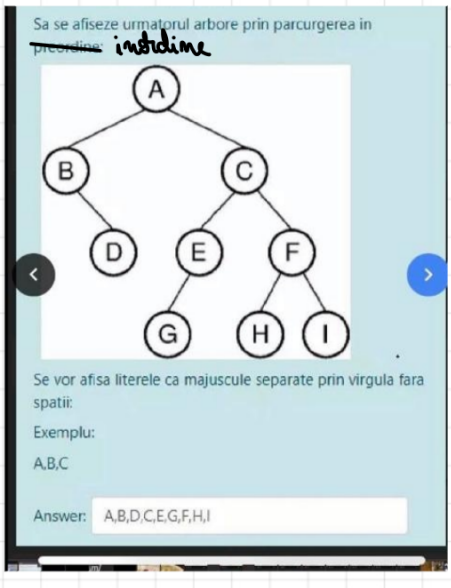
Y - nu are voie să aibă frunze





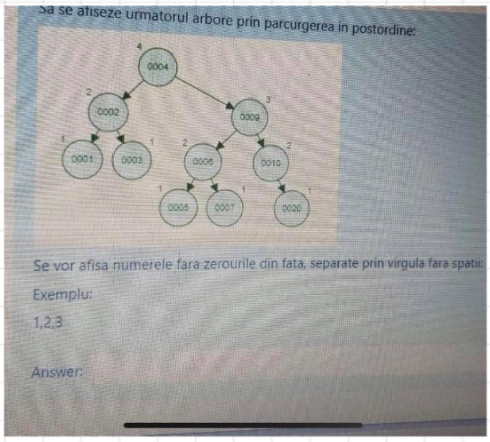
postordine (SBR)

B, B, G, E, H, I, F, C, A



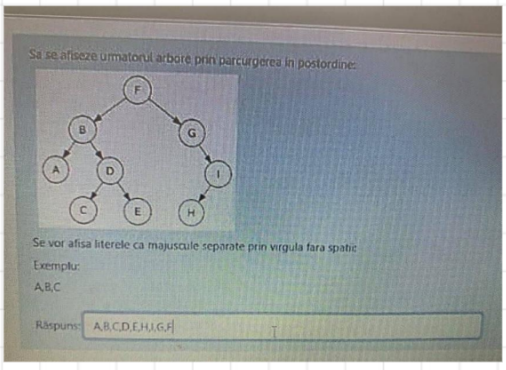
SRD

B, B, A, G, E, H, F, I, C



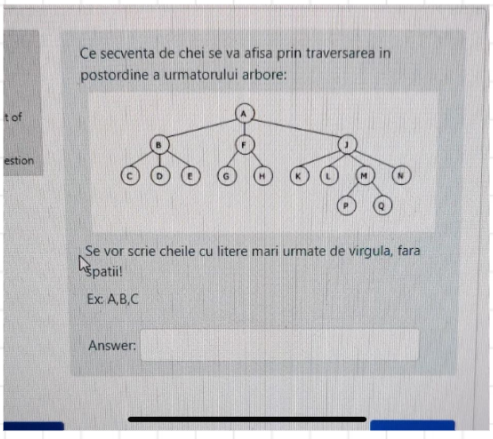
SBR

1, 3, 2, 5, 7, 6, 20, 10, 9, 4



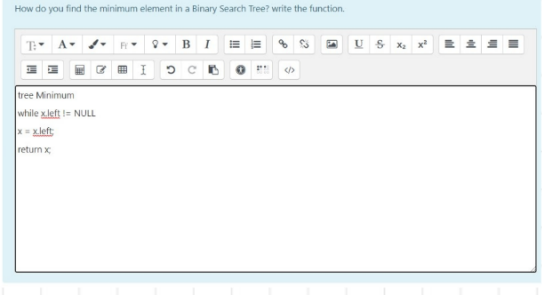
SBR

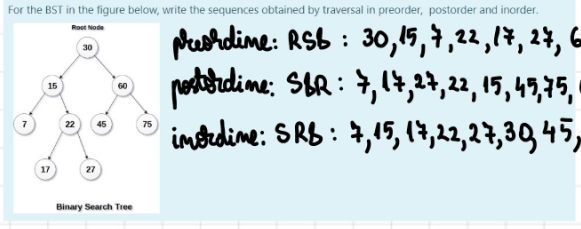
A, C, E, B, B, H, I, G, F



SBR

C, B, E, B, G, H, F, K, L, P, Q, M, N, J, A

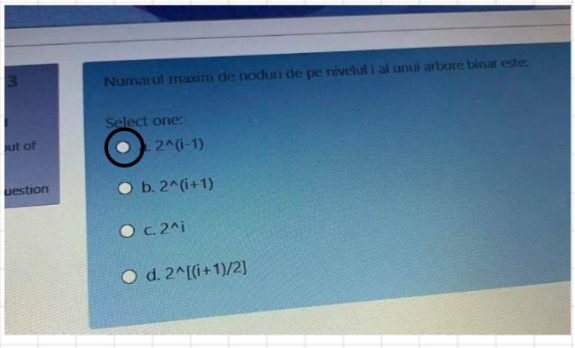




preordine: RSB : 30, 15, 7, 22, 17, 27, 60, 45, 75

postordine: SBR : 7, 17, 27, 22, 15, 45, 75, 60, 30

inordine: SRB : 7, 15, 17, 22, 27, 30, 45, 60, 75



1: 2^0
2: 2^1
3: 2^2

$\Big\} 2^{i-1}$

Un nod oarecare al unui arbore binar are:

Select one:

a. oricati fii

b. exact doi fii

c. cel mult doi fii

d. cel putin doi fii

Numarul maxim de noduri intr-un arbore binar de inaltime h este:

Select one:

a. h^2

b. 2^h -1

c. (h + 1) ^ 2

d. 2^(h -1)

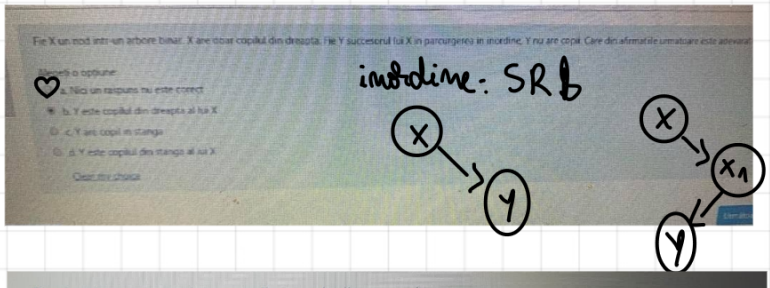
e. 2^(h+1) -1

Clear my choice

1: $2^1 - 1$

2: $2^2 - 1$

3: $2^3 - 1$ (7)



Pentru traversarea in preordine nodurile se prelucreaza:

Select one:

a. a doua oara cand sunt intalnite

b. prima data cand sunt intalnite

c. se prelucreaza un nod terminal la prima intalnire si unul interior la a doua intalnire

d. ultima data cand sunt intalnite

R Sb

oritmilor

Question 8

Not yet answered

Marked out of 1.00

Flag question

Pentru traversarea in inordine nodurile se prelucreaza:

S R b

Select one:

a. prima data cand sunt intalnite

b. ultima data cand sunt intalnite

pre

post

c. se prelucreaza un nod terminal la prima intalnire si unul interior la a doua intalnire

d. a doua oara cand sunt intalnite

Previous page

Arbori AVL

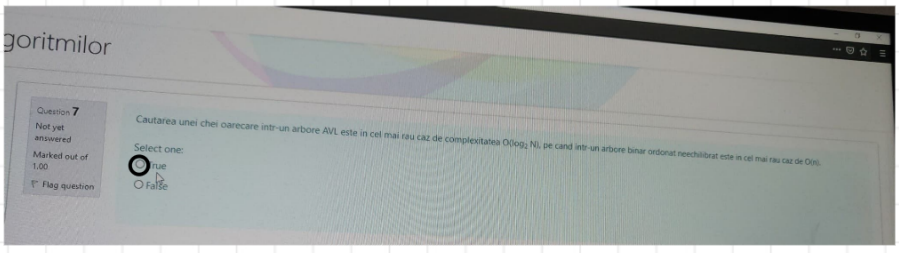
What is the maximum possible height of an AVL tree with N nodes?

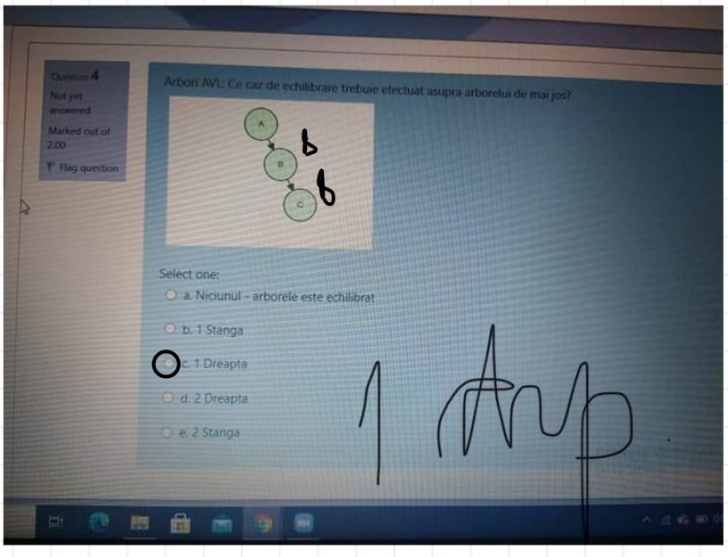
the maximum possible height of an AVL tree with n nodes is $1.44 \log_2 (n)$

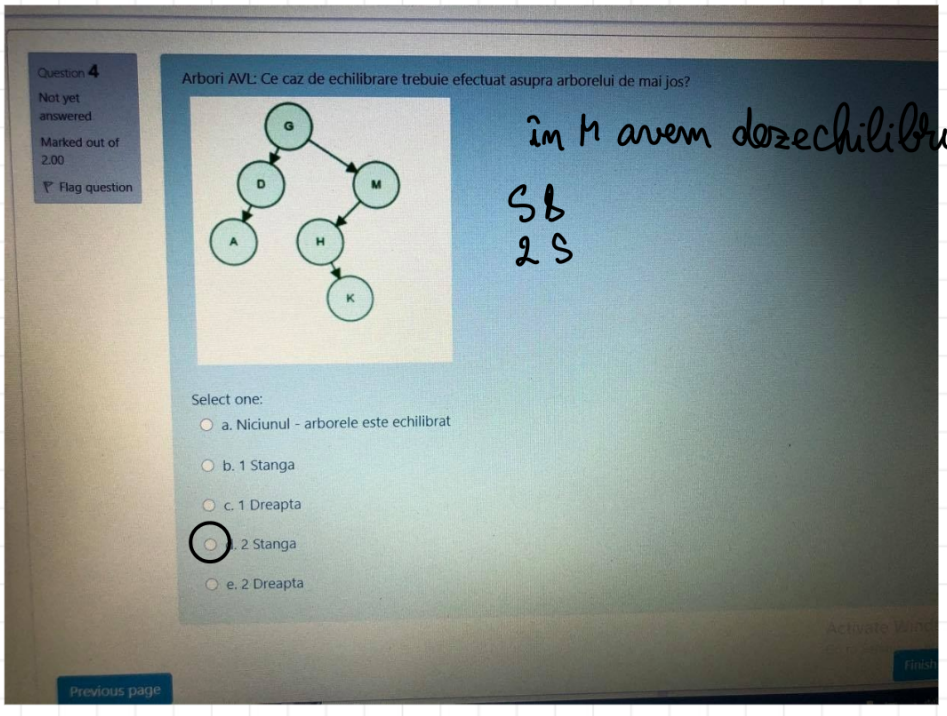
aparent e bine

1 Explain in which conditions a double rotation having the directions left-right is used in AVL trees.

2S Sb -> pe curs



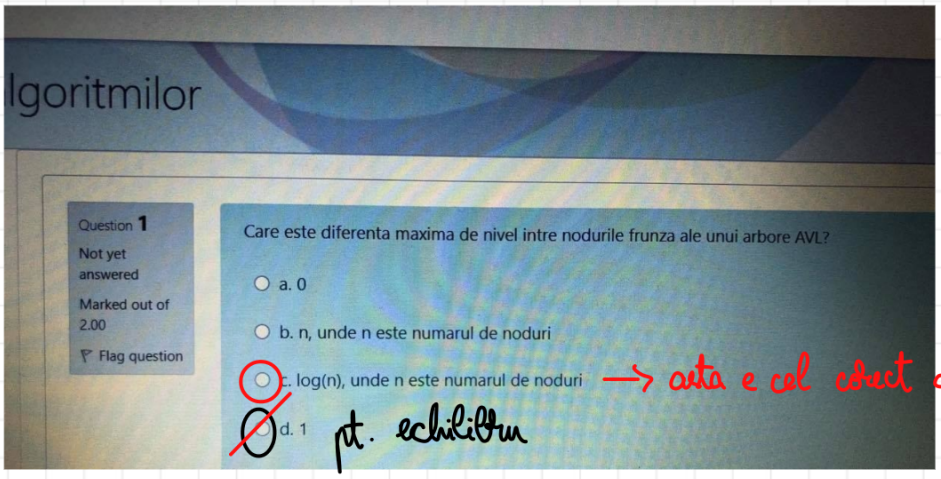


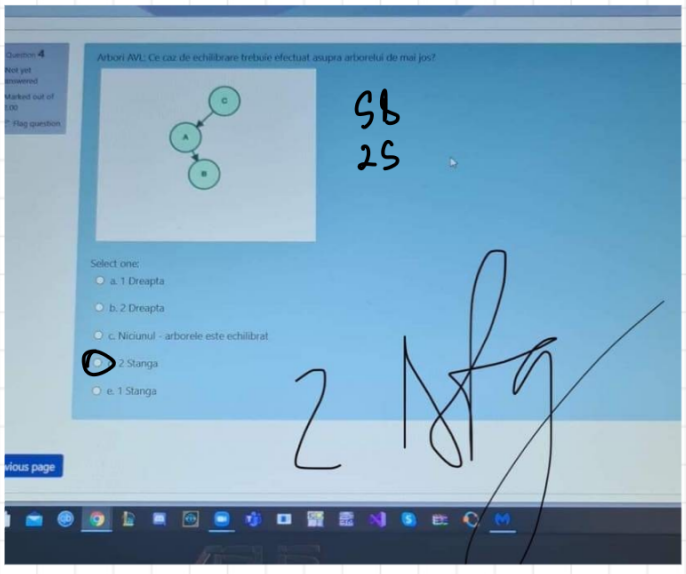


în M avem dezechilibru

Sb

2S





Question 2
Answer saved
Marked out of 3.00
Flag question

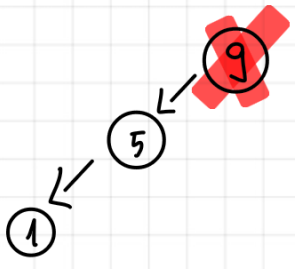
Sa se scrie configuratia pe niveluri a arborelui AVL rezultat dupa insertia intr-un arbore vid a cheilor urmatoare in ordinea data mai jos:

9,5,1,2,3

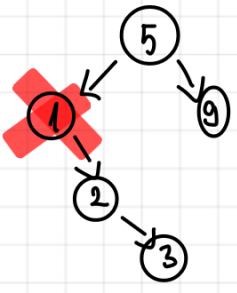
Pentru fiecare nivel (de sus in jos) se vor scrie cheile (de la stanga la dreapta) separate prin virgula (,), iar nivelurile vor fi separate prin bara verticala (|) fara spatii.

Exemplu A|B,C|D,E,F,G - reprezinta un arbore cu nodul A pe primul nivel, nodurile B si C pe al doilea nivel si D, E, F, G pe al treilea nivel.

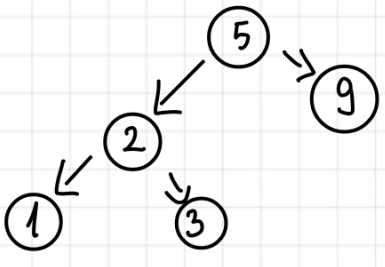
Answer: 5|2,9|1,3



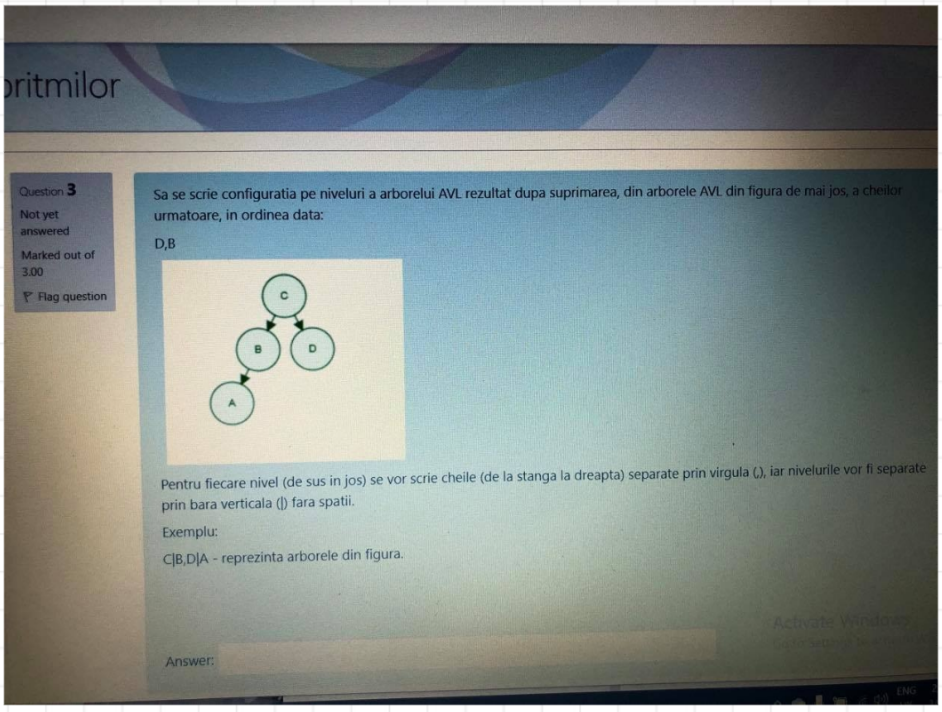
car 1 S → SS



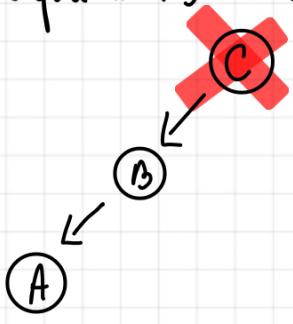
car 1 D → DD



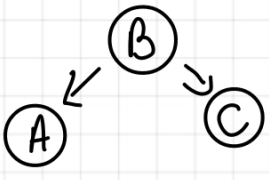
=> 5|2,9|1,3



Ștergem B -> frunză

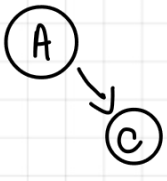


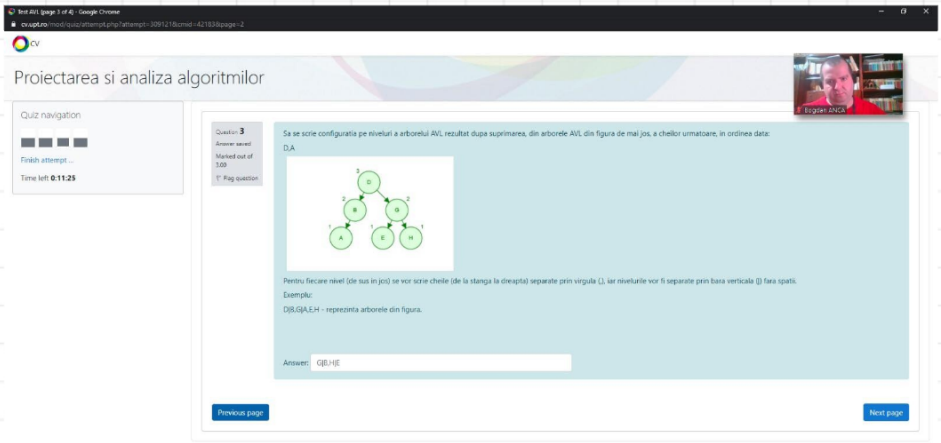
dezechilibru în C , caz 1S (SS)



parcurgere inordine A, B, C

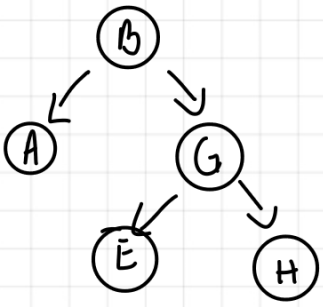
Ștergem B -> predecesor inordine: A



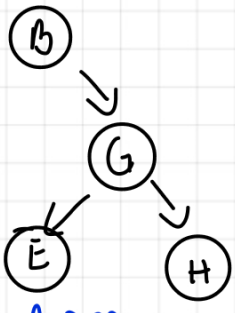


Parcurgere inordine: A, B, B, E, G, H

Suprimăm B -> predecesor inordine: B



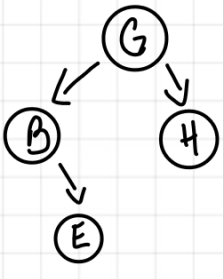
Suprimăm A -> frunză



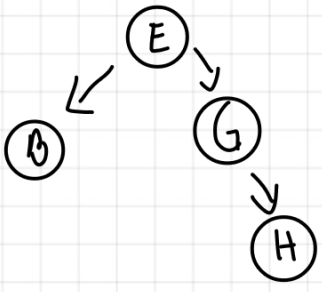
Dezechilibru în B -> avem 2 cazuri → 1B (BB)
→ 2B (BS)

1B leg cu E se taie

2B → E cu B



G | B,H | E



E | B,G | H

Test AVL (page 2 of 4) - Google Chrome

cv.upt.ro/mod/quiz/attempt.php?attempt=309121&cmid=42183&page=1

cv

Proiectarea si analiza algoritmilor

Quiz navigation

Finish attempt ...

Time left 0:11:35

Question 2

Answer saved

Marked out of 3.00

Flag question

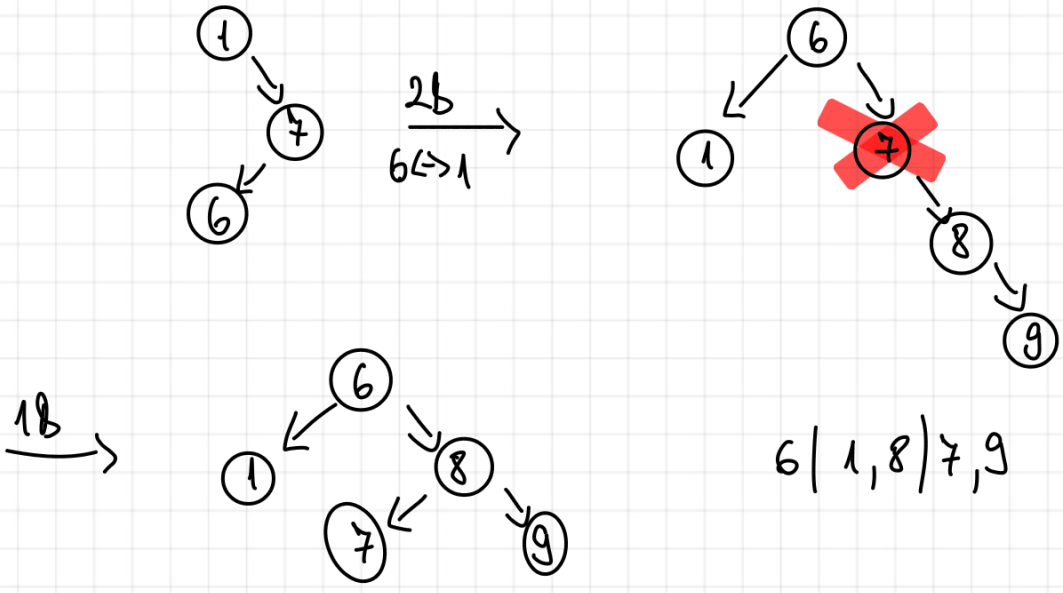
Sa se scrie configuratia pe niveluri a arborelui AVL rezultat dupa insertia intr-un arbore vid a cheilor urmatoare in ordinea data mai jos:

1,7,6,8,9

Pentru fiecare nivel (de sus in jos) se vor scrie cheile (de la stanga la dreapta) separate prin virgula (,), iar nivelurile vor fi separate prin bara verticala (|) fara spatii.

Exemplu A|B,C|D,E,F,G - reprezinta un arbore cu nodul A pe primul nivel, nodurile B si C pe al doilea nivel si D, E, F, G pe al treilea nivel.

Answer: 6|1,8|7,9

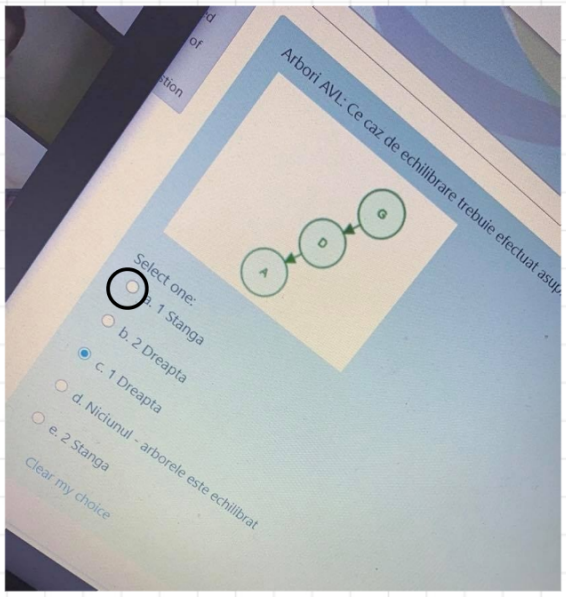


Arborii AVL se comporta la fel ca arborii binari ordonati simpli, mai putin in cazul operatiilor d

insertie si suprimare de chei

calcul al inaltimii

initializare



SS

1S

Intr-un arbore AVL toate nodurile frunza se gasesc la aceeasi adancime.

Select one:

True

False

m sau (m-1)

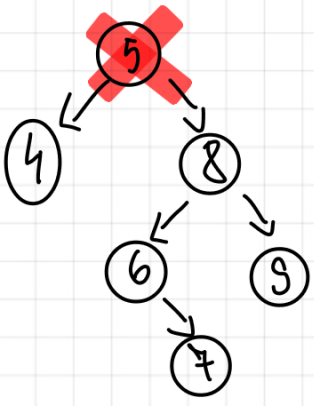
Sa se scrie configuratia pe niveluri a arborelui AVL rezultat dupa insertia intr-un arbore vid a cheilor urmatoare in ordinea data mai jos:

5,4,8,6,9,7

Pentru fiecare nivel (de sus in jos) se vor scrie cheile (de la stanga la dreapta) separate prin virgula (,), iar nivelurile vor fi separate prin bara verticala (|) fara spatii.

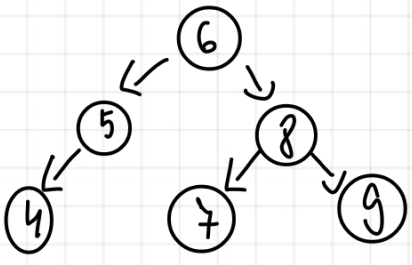
Exemplu A|B,C|D,E,F,G - reprezinta un arbore cu nodul A pe primul nivel, nodurile B si C pe al doilea nivel si D, E, F, G pe al treilea nivel.

Answer:



Dezechilibru în 5 caz 2b RS

5 ↔ 6 tăiem legătura cu 7



6 | 5,8 | 4,7,9